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Die assembly and method of forming a vinyl record

Abstract

A die assembly and a method of forming a vinyl record are provided. The assembly includes a base circular-shaped plate having an outer ring-shaped gasket groove, an inner O-ring groove, and a plurality of concentric grooves extending from a primary surface into the base circular-shaped plate. The assembly includes an O-ring that is disposed in the inner O-ring groove, and an outer ring-shaped gasket disposed in the outer ring-shaped gasket groove. The assembly includes a contacting circular-shaped plate having an engagement surface and a plurality of concentric channels such that a plurality of concentric finger portions are formed on the contacting circular-shaped plate. The plurality of concentric finger portions are partially disposed in the plurality of concentric grooves and contact the base circular-shaped plate.

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Background/Summary

BACKGROUND

(1) Die assemblies have been utilized to form vinyl records. The die assemblies have metal bonded plates which are inseparable after manufacturing without damaging the metal bonded plates. Accordingly, manufactures of vinyl records are not able to separate the metal bonded plates apart from one another for maintenance purposes and therefore throw the die assemblies away when the die assemblies have impaired functionality such as leaking fluid therefrom.

(2) The inventor herein has recognized a need for an improved die assembly and method for forming a vinyl record that minimizes and/or reduces the above-mentioned problem.

SUMMARY

(3) A die assembly in accordance with an exemplary embodiment is provided. The die assembly includes a base circular-shaped plate having a primary surface and an outer ring-shaped gasket groove, an inner O-ring groove, and a plurality of concentric grooves extending from the primary surface into the base circular-shaped plate. The plurality of concentric grooves are disposed between the outer ring-shaped gasket groove and the inner O-ring groove. The base circular-shaped plate has an inlet aperture extending from an outer peripheral surface thereof into the base circular-shaped plate to a central region of the base circular-shaped plate. The inlet aperture communicates with a first vertical aperture that extends from an end of the inlet aperture and through the primary surface. The base circular-shaped plate has an outlet aperture extending from the outer peripheral surface into the base circular-shaped plate. The outlet aperture communicates with a second vertical aperture that extends from an end of the outlet aperture and through the primary surface. The base circular-shaped plate has a plurality of peripheral bolt apertures extending therethrough that are disposed radially outwardly from the outer ring-shaped gasket groove. The die assembly further includes an outer ring-shaped gasket disposed in the outer ring-shaped gasket groove. The die assembly further includes an O-ring that is disposed in the inner O-ring groove. The die assembly further includes a contacting circular-shaped plate having an engagement surface and a plurality of concentric channels extending from the engagement surface into the contacting circular-shaped plate such that a plurality of concentric finger portions are formed on the contacting circular-shaped plate. Each concentric finger portion is disposed between two concentric channels of the plurality of concentric channels. The contacting circular-shaped plate has a plurality of radial passages

fluidly interconnecting the plurality of concentric channels. The contacting circular-shaped plate is disposed on the base circular-shaped plate such that the plurality of concentric finger portions are partially disposed in the plurality of concentric grooves and contact the base circular-shaped plate. The inlet aperture fluidly communicates with the plurality of concentric channels. The outlet aperture fluidly communicates with the plurality of concentric channels. A flow path is formed by and through the inlet aperture, the first vertical aperture, the plurality of concentric channels, the plurality of radial passages, the second vertical aperture, and the outlet aperture. The contacting circular-shaped plate has a plurality of peripheral bolt holes extending therein that are disposed radially outwardly from the plurality of concentric channels. The die assembly further includes a plurality of bolts extending through the plurality of peripheral bolt apertures of the base circular-shaped plate and further into the plurality of peripheral bolt holes of the contacting circular-shaped plate to couple the base circular-shaped plate to the contacting circular-shaped plate.

(4) A method of forming a vinyl record in accordance with another exemplary embodiment is provided. The method includes providing a vinyl record manufacturing system having a first die assembly, a second die assembly, a first stamper plate coupled to the first die assembly, a second stamper plate coupled to the second die assembly, a heating system, and a least one actuator. The first die assembly includes a base circular-shaped plate having a primary surface and an outer ring-shaped gasket groove, an inner O-ring groove, and a plurality of concentric grooves extending from the primary surface into the base circular-shaped plate. The plurality of concentric grooves are disposed between the outer ring-shaped gasket groove and the inner O-ring groove. The base circular-shaped plate has an inlet aperture extending from an outer peripheral surface thereof into the base circular-shaped plate to a central region of the base circular-shaped plate. The inlet aperture communicates with a first vertical aperture that extends from an end of the inlet aperture and through the primary surface. The base circular-shaped plate has an outlet aperture extending from the outer peripheral surface into the base circular-shaped plate. The outlet aperture communicates with a second vertical aperture that extends from an end of the outlet aperture and through the primary surface. The base circular-shaped plate has a plurality of peripheral bolt apertures extending therethrough that are disposed radially outwardly from the outer ring-shaped gasket groove. The first die assembly further includes an outer ring-shaped gasket disposed in the outer ring-shaped gasket groove. The first die assembly further includes an O-ring that is disposed in the inner O-ring groove. The first die assembly further includes a contacting circular-shaped plate having an engagement surface and a plurality of concentric channels extending from the engagement surface into the contacting circular-shaped plate such that a plurality of concentric finger portions are formed on the contacting circular-shaped plate. Each concentric finger portion is disposed between two concentric channels of the plurality of concentric channels. The contacting circular-shaped plate has a plurality of radial passages fluidly interconnecting the plurality of concentric channels. The contacting circular-shaped plate is disposed on the base circular-shaped plate such that the plurality of concentric finger portions are partially disposed in the plurality of concentric grooves and contact the base circular-shaped plate. The inlet aperture fluidly communicates with the plurality of concentric channels. The outlet aperture fluidly communicates with the plurality of concentric channels. A flow path is formed by and through the inlet aperture, the first vertical aperture, the plurality of concentric channels, the plurality of radial passages, the second vertical aperture, and the outlet aperture. The contacting circular-shaped plate has a plurality of peripheral bolt holes extending therein that are disposed radially outwardly from the plurality of concentric channels. The first die assembly further includes a plurality of bolts extending through the plurality of peripheral bolt apertures of the base circular-shaped plate and further into the plurality of peripheral bolt holes of the contacting circular-shaped plate to couple the base circular-shaped plate to the contacting circular-shaped plate. The method further includes disposing a polymeric puck between the first and second stamper plates. The method further includes heating the first and second die assemblies by pumping steam through the flow path of the

first die assembly and through a flow path in the second die assembly. The method further includes moving the first and second die assemblies toward one another such that the first and second stamper plates compress the polymeric puck to form the vinyl record.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic of a vinyl record manufacturing system having first and second die assemblies, first and second stamper plates, a heating system, a cooling system, an actuator, and a machine controller;

(2) FIG. **2** is a side view of the first and second dies assemblies and the first and second stamper plates of FIG. **1** and a polymeric puck prior to the puck being compressed;

(3) FIG. **3** is a side view of the first and second dies assemblies and the first and second stamper plates of FIG. **2** and the polymeric puck after the puck is compressed into a vinyl record;

(4) FIG. **4** is an isometric view of a vinyl record;

(5) FIG. **5** is an exploded view of the first die assembly and the first stamper plate of FIG. **1**;

(6) FIG. **6** is an isometric view of the first die assembly of FIG. **1**;

(7) FIG. **7** is another isometric view of the first die assembly of FIG. **6**;

(8) FIG. **8** is a top view of the first die assembly of FIG. **6**;

(9) FIG. **9** is a bottom view of first die assembly of FIG. **1**;

(10) FIG. **10** is another isometric view of the first die assembly of FIG. **6**;

(11) FIG. **11** is a side view of the first die assembly of FIG. **6**;

(12) FIG. **12** is another side view of the first die assembly of FIG. **6**;

(13) FIG. **13** is another side view of the first die assembly of FIG. **6**;

(14) FIG. **14** is a cross-sectional view of the first die assembly of FIG. **13** taken along lines **14-14** in FIG. **13**;

(15) FIG. **15** is an enlarged view of a portion of the first die assembly of FIG. **14**;

(16) FIG. **16** is an enlarged view of another portion of the first die assembly of FIG. **14**;

(17) FIG. **17** is a cross-sectional view of the first die assembly of FIG. **3** taken along lines **17-17** in FIG. **3**;

(18) FIG. **18** is a partial cut-through view of the first die assembly of FIG. **1** taken along lines **18-18** in FIG. **1**;

(19) FIG. **19** is an isometric view of a base circular-shaped plate utilized in the first die assembly of FIG. **6**;

(20) FIG. **20** is a partial transparent view of the base circular-shaped plate of FIG. **19**;

(21) FIG. **21** is a top view of the base circular-shaped plate of FIG. **19**;

(22) FIG. **22** is an isometric view of the base circular-shaped plate of FIG. **19** and a plurality of peripheral bolts utilizing therein;

(23) FIG. **23** is an isometric view of a contacting circular-shaped plate utilized in the first die assembly of FIG. **6**;

(24) FIG. **24** is another isometric view of the contacting circular-shaped plate of FIG. **23**;

(25) FIG. **25** is a side view of the contacting circular-shaped plate of FIG. **23**;

(26) FIG. **26** is another side view of the contacting circular-shaped plate of FIG. **23**;

(27) FIG. **27** is a top view of the contacting circular-shaped plate of FIG. **23**;

(28) FIG. **28** is another isometric view of the contacting circular-shaped plate of FIG. **23**;

(29) FIG. **29** is a bottom view of the contacting circular-shaped plate of FIG. **23**;

(30) FIG. **30** is a partially transparent view of the contacting circular-shaped plate of FIG. **23** disposed on the base circular-shaped plate of FIG. **19**;

(31) FIG. **31** is a top view of a first stamper plate utilized in the vinyl record manufacturing system

of FIG. **1**;
(32) FIG. **32** is a bottom view of the first stamper plate of FIG. **31**;
(33) FIG. **33** is a side view of the first stamper plate of FIG. **31**;
(34) FIG. **34** is a top view of a second stamper plate utilized in the vinyl record manufacturing system of FIG. **1**; and
(35) FIGS. **35-36** are flowcharts of a method of forming a vinyl record utilizing the vinyl record manufacturing system in accordance with another exemplary embodiment.

DETAILED DESCRIPTION

(36) Referring to FIGS. **1-4**, a vinyl record manufacturing system **20** in accordance with an exemplary embodiment will be explained. The vinyl record manufacturing system **20** produces a vinyl record **28** (shown in FIG. **4**) utilized the polymeric puck **24** (shown in FIG. **2**).
(37) Referring to FIG. **1**, the vinyl record manufacturing system **20** includes a first die assembly **51**, a second die assembly **52**, a first stamper plate **61**, a second stamper plate **62**, a heating system **66**, a cooling system **70**, an actuator **81**, and a machine controller **86**.
(38) Referring to FIGS. **5-31**, the first die assembly **51** is provided to hold the first stamper plate **61** thereon and to heat and cool the first stamper plate **61**. Referring to FIGS. **5** and **14-31**, the first die assembly **51** includes a base circular-shaped plate **100**, an outer ring-shaped gasket **104**, an O-ring **108**, a contacting circular-shaped plate **112**, a threaded bolt **116** (shown in FIG. **17**), a center bushing **118** (shown in FIG. **5**), and a nut **120**.
(39) Referring to FIG. **5**, an advantage of the first die assembly **51** is that the assembly **51** utilizes a base circular-shaped plate **100** having a plurality of concentric grooves **160** receiving at least a portion of a plurality of concentric finger portions **420** of the contacting circular-shaped plate **112** therein to partition portions of the assembly **51** for routing either steam or a cooled fluid therethrough. Referring to FIG. **19**, another advantage of the first die assembly is that the assembly **51** utilizes an outer ring-shaped gasket **104** and an O-ring **108** to maintain fluid within desired regions of the assembly **51**. Still further, the first die assembly **51** has an advantage of utilizing a plurality of bolts **122** to couple together the base circular-shaped plate **100** and the contacting circular-shaped plate **112** without the plates **100**, **112** being welded together, metal bonded together, or permanently bonded together using another means such as an adhesive or epoxy. As a result, a user can desirably separate the plates **100**, **112** to perform maintenance on the plates **100**, **112**.
(40) Referring to FIGS. **14-18**, the base circular-shaped plate **100** is disposed on and coupled to the contacting circular-shaped plate **112**. The base circular-shaped plate **100** includes a primary surface **140**, an outer peripheral surface **144**, a back surface **148**, an outer ring-shaped gasket groove **152**, an inner O-ring groove **156**, a plurality concentric grooves **160**, a ring-shaped placement groove **164**, a center aperture **166**, an inlet aperture **168** (shown in FIGS. **14** and **18**), a first vertical aperture **172** (shown in FIG. **14**), an outlet aperture **176** (shown in FIG. **20**), a second vertical aperture **180**, a central region **184** (shown in FIG. **19**), a plurality of peripheral apertures **188** (shown in FIGS. **9** and **10**), first, second, third, and fourth clamping grooves **201**, **202**, **203**, **204** (shown in FIGS. **11-13** and **19**), and a plurality of inner bolt apertures **212** (shown in FIGS. **9** and **10**). In an exemplary embodiment, the base circular-shaped plate **100** is constructed of steel.
(41) Referring to FIG. **14**, the primary surface **140** and the back surface **148** are disposed opposite to and parallel to one another. Further, the outer peripheral surface **144** extends perpendicular to and between the primary surface **140** and the back surface **148**.
(42) Referring to FIGS. **14** and **19**, the outer ring-shaped gasket groove **152** extends from the primary surface **140** into the base circular-shaped plate **100**. The outer ring-shaped gasket groove **152** is provided to hold the outer ring-shaped gasket **104** therein. A radial width of the outer ring-shaped gasket groove **144** is greater than a radial width of the inner O-ring groove **156**.
(43) The inner O-ring groove **156** is disposed in the central region **184** of the base circular-shaped plate **100** and extends from the primary surface **140** into the base circular-shaped plate **100**. The inner O-ring groove **156** is provided to hold the O-ring **108** therein.

(44) Referring to FIGS. **16** and **21**, the plurality of concentric grooves **160** extend from the primary surface **140** into the base circular-shaped plate **100**. The plurality of concentric grooves **160** are disposed between the outer ring-shaped gasket groove **152** and the inner O-ring groove **156**. The plurality concentric grooves **160** include concentric grooves **221**, **222**, **223**, **224**, **225**, **226**, **227**, **228**, **229**, **230**, **231**, **232**.

(45) Referring to FIG. **14**, the ring-shaped placement groove **164** is provided to position the contacting circular-shaped plate **112** on the base circular-shaped plate **100**. The ring-shaped placement groove **164** extends from the primary surface **140** into the base circular-shaped plate **100**. The ring-shaped placement groove **164** extends concentrically around the outer ring-shaped gasket groove **152**. A depth of the ring-shaped placement groove **164** is greater than a depth of the outer ring-shaped gasket groove **152**. Also, the depth of the ring-shaped placement groove **164** is greater than a depth of the plurality of concentric grooves **160**. Further, a radial width of the ring-shaped placement groove **164** is greater than a radial width of the outer ring-shaped gasket groove **152**.

(46) Referring to FIGS. **14**, **19** and **20**, the center aperture **166** is a threaded aperture and extends through the base circular-shaped plate **100** from the primary surface **140** to the back surface **148**. The center aperture **166** is provided to receive the threaded bolt **116** therein to secure the base circular-shaped plate **100** to the contacting circular-shaped plate **112**.

(47) Referring to FIGS. **14** and **18**, the inlet aperture **168** extends from an outer peripheral surface **144** into the base circular-shaped plate **100** to a central region **184** of the base circular-shaped plate **100**. The inlet aperture **168** communicates with the first vertical aperture **172** that extends through the central region **184** from an end of the inlet aperture **168** and through the primary surface **140**. The inlet aperture **168** fluidly communicates with the plurality of concentric channels **416**.

(48) Referring to FIG. **20**, the outlet aperture **176** extends from the outer peripheral surface **144** into the base circular-shaped plate **100**. The outlet aperture **176** communicates with a second vertical aperture **180** that extends from an end of the outlet aperture **176** and through the primary surface **140**. The outlet aperture **176** fluidly communicates with the plurality of concentric channels **416**. The second vertical aperture **180** communicates with the radially outermost concentric channel of the plurality of concentric channels **416**.

(49) Referring to FIGS. **15**, **16** and **19**, the central region **184** is disposed between the inner O-ring groove **156** and the radially concentric groove **222** of the plurality of concentric grooves **160**.

(50) Referring to FIGS. **9**, **10** and **21**, the plurality of peripheral bolt apertures **188** extend through the base circular-shaped plate **100**. The plurality of peripheral bolt apertures **188** are disposed radially outwardly from the outer ring-shaped gasket groove **152** (shown in FIG. **14**). The plurality of peripheral bolt apertures **188** include peripheral bolt apertures **251**, **252**, **253**, **254**, **255**, **256**, **257**, **258**, **259**, **260**, **261**, **262**, **263**, **264**, **265**, **266**, **267**, **268**, **269**, **270**, **271**, **272**, **273**.

(51) Referring to FIGS. **11-13** and **19**, the first, second, third, and fourth clamping grooves **201**, **202**, **203**, **204** are disposed from the outer peripheral surface **144** into the base circular-shaped plate **100**.

(52) Referring to FIGS. **9** and **10**, the plurality of inner bolt apertures **212** extend from the back surface **148** into the base circular-shaped plate **100**. The plurality of inner bolt apertures **212** are provided to allow the base circular-shaped plate **100** to be coupled to the actuator **81** (shown in FIG. **1**). The plurality of inner bolt apertures **212** include inner bolt apertures **331**, **332**, **333**, **334**, **335**, **336**, **337**, **338**, **339**, **340**, **341**, **342**.

(53) Referring to FIGS. **15** and **23-31**, the contacting circular-shaped plate **112** is provided to be coupled to the base circular-shaped plate **100**. The contacting circular-shaped plate **112** includes an engagement surface **400** (shown in FIG. **15**), an outer peripheral surface **404**, a stamper plate holding surface **408**, a ring-shaped placement member **412**, a plurality of concentric channels **416**, a plurality of concentric finger portions **420**, a plurality of radial passages **424** (shown in FIGS. **29** and **30**), a plurality of peripheral bolt holes **428** (shown in FIG. **28**), a center aperture **432**, and first,

second, third, and fourth clamping grooves **441**, **442**, **443**, **444**. In an exemplary embodiment, the contacting circular-shaped plate **112** is constructed of steel.

(54) Referring to FIGS. **15**, **24** and **29**, the engagement surface **400** and the stamper plate holding surface **408** are disposed opposite to and substantially parallel to one another. Further, the outer peripheral surface **404** extends perpendicular to and between the engagement surface **400** and the stamper plate holding surface **408**.

(55) Referring to FIGS. **14**, **15** and **28**, the ring-shaped placement member **412** extends outwardly from the engagement surface **400** and around an outer periphery of the engagement surface **400**. The ring-shaped placement member **412** is received in the ring-shaped placement groove **164** of the base circular-shaped plate **100**. The ring-shaped placement member **412** has first and second cut-out regions **451**, **452** extending therethrough.

(56) Referring to FIGS. **15**, **16**, **28** and **29**, the plurality of concentric channels **416** extend from the engagement surface **400** into the contacting circular-shaped plate **112** such that a plurality of concentric finger portions **420** (shown in FIG. **15**) are formed on the contacting circular-shaped plate **112**. The plurality of concentric channels **416** includes concentric channels **461**, **462**, **463**, **464**, **465**, **466**, **467**, **468**, **469**, **470**, **471**, **472**. The plurality of concentric finger portions **420** includes concentric finger portions **491**, **492**, **493**, **494**, **495**, **496**, **497**, **498**, **499**, **500**, **501**, **502**. Each of the concentric finger portions **492**, **493**, **494**, **495**, **496**, **497**, **498**, **499**, **500**, **501**, **502** are disposed between two concentric channels of the plurality of concentric channels **416**. The concentric finger portions **491**, **492**, **493**, **494**, **495**, **496**, **497**, **498**, **499**, **500**, **501**, **502** of the contacting circular-shaped plate **100** are disposed in the concentric grooves **221**, **222**, **223**, **224**, **225**, **226**, **227**, **228**, **229**, **230**, **231**, **232**, respectively in the base circular-shaped plate **100**.

(57) Referring to FIGS. **28** and **29**, the plurality of radial passages **424** extend from the engagement surface **400** (shown in FIG. **15**) into the contacting circular-shaped plate **112**. The plurality radial passages **424** includes radial passages **531**, **532**, **533**, **534**, **535**, **536**, **537**, **538**, **539**, **540**, **541**, **542**, **543**, **544**, **545**, **546**, **547**, **548**, **549**, **550**, **551**, **552** which fluidly interconnecting the plurality of concentric channels **416**.

(58) As discussed briefly above, referring to FIGS. **15**, **16**, **18**, **28** and **29**, the contacting circular-shaped plate **112** is disposed on the base circular-shaped plate **100** such that the plurality of concentric finger portions **420** are partially disposed in the plurality of concentric grooves **160** and contact the base circular-shaped plate **100**, and a flow path is formed by and through the inlet aperture **168**, the first vertical aperture **172**, the plurality of concentric channels **416**, the plurality of radial passages **424**, the second vertical aperture **180**, and the outlet aperture **176**. During operation, when heating the first die assembly **51**, steam flows through the flow path to heat the first die assembly **51**. Further, when cooling the first die assembly **51**, a cooling fluid flows through the flow path to cool the first die assembly **51**.

(59) Referring to FIGS. **22**, **28** and **29**, the plurality of peripheral bolt holes **428** extend from the engagement surface **400** (shown in FIG. **14**) into the contacting circular-shaped plate **112**. The plurality of peripheral bolt holes **428** are disposed radially outwardly from the plurality of concentric channels **416**. The plurality of peripheral bolt holes **428** include peripheral bolt holes **571**, **572**, **573**, **574**, **575**, **576**, **577**, **578**, **579**, **580**, **581**, **582**, **583**, **584**, **585**, **586**, **537**, **588**, **589**, **590**, **591**, **592**, **593**. The bolts **291**, **292**, **293**, **294**, **295**, **296**, **297**, **298**, **299**, **300**, **301**, **302**, **303**, **304**, **305**, **306**, **307**, **308**, **309**, **310**, **311**, **312**, **313** extend through the peripheral bolt hole apertures **251**, **252**, **253**, **254**, **255**, **256**, **257**, **258**, **259**, **260**, **261**, **262**, **263**, **264**, **265**, **266**, **267**, **268**, **269**, **270**, **271**, **272**, **273**, respectively, of the base circular-shaped plate **100** and into the peripheral bolt holes **571**, **572**, **573**, **574**, **575**, **576**, **577**, **578**, **579**, **580**, **581**, **582**, **583**, **584**, **585**, **586**, **537**, **588**, **589**, **590**, **591**, **592**, **593**, respectively, of the contacting circular-shaped plate **112**—to couple the base circular-shaped plate **100** to the contacting circular-shaped plate **112**.

(60) Referring to FIG. **29**, the center aperture **432** extends through the contacting circular-shaped plate **112**. The center aperture **432** aligns with the center aperture **166** (shown in FIG. **19**) of the

base circular-shaped plate **100** when the contacting circular-shaped plate **112** is disposed on the base circular-shaped plate **100**. Referring to FIG. **14**, the threaded bolt **116** is threadably received through the center apertures **432**, **166** to couple the base circular-shaped plate **100** to the contacting circular-shaped plate **112**.

(61) Referring to FIGS. **23-26**, the outer peripheral surface **404** of the contacting circular-shaped plate **112** has first, second, third, and fourth clamping grooves **441**, **442**, **443**, **444** disposed therein. The first, second, third, and fourth clamping grooves **441**, **442**, **443**, **444** align with the first, second, third, and fourth clamping grooves **201**, **202**, **203**, **204** (shown in FIGS. **11-13**) when the base circular-shaped plate **100** is coupled to the contacting circular-shaped plate **112**.

(62) Referring to FIGS. **14** and **21**, the O-ring **108** is disposed in the inner O-ring groove **156** of the base circular-shaped plate **100**. Further, the O-ring **108** is compressed by the finger portion **221** (shown in FIG. **16**) when the contacting circular-shaped plate **112** is disposed on the base circular-shaped plate **100**.

(63) The outer ring-shaped gasket **104** is disposed in the outer ring-shaped gasket groove **152** of the base circular-shaped plate **100**. Further, the outer ring-shaped gasket **104** is compressed by the contacting circular-shaped plate **112** when the contacting circular-shaped plate **112** is disposed on the base circular-shaped plate **100**.

(64) Referring to FIGS. **5** and **14**, the center bushing **118** extends through the center aperture **638** of the first stamper plate **61**, the center aperture **600** of the threaded bolt **116**, and the center aperture **166** of the base circular-shaped plate **100** and is threadably coupled to the nut **120** to couple the first stamper plate **61** to the contacting circular-shaped plate **112**. Alternately, the base circular-shaped plate **100** is coupled to the contacting circular-shaped plate **112** utilizing a slotted bar that is integrally coupled to the system **20**.

(65) Referring to FIGS. **1**, **4** and **31-33**, the first stamper plate **61** is removably coupled to the first die assembly **51**. The first stamper plate **61** includes a polymeric puck contacting surface **630** and a die assembly contacting surface **634**. The polymeric puck contacting surface **630** is disposed opposite to the die assembly contacting surface **634**. The first stamper plate **61** further includes a center aperture **638** extending therethrough. The polymeric puck contacting surface **630** has ridges disposed thereon corresponding to a musical composition that is imprinted on a first side of the vinyl record **28**. In an exemplary embodiment, the first stamper plate **61** is constructed of a coated metal.

(66) Referring to FIGS. **1-3**, the second die assembly **52** has an identical structure as the first die assembly **51**. The second die assembly **52** includes an inlet aperture **652** and an outlet aperture **654**.

(67) Referring to FIGS. **1** and **34**, the second stamper plate **62** is provided to be removably coupled to the second die assembly **52**. The second stamper plate **62** has a substantially structure as the first stamper plate **61**. The only difference between the second stamper plate **62** and the first stamper plate **61** is that the second stamper plate **62** has a polymeric puck contacting surface having ridges disposed thereon corresponding to another musical composition that is imprinted on a second side of the vinyl record **28**. The second stamper plate **62** is removably coupled to the second die assembly **52** in an identical fashion as the first stamper plate **61** is removably coupled to the first die assembly **51**.

(68) Referring to FIG. **1**, the remaining components of the vinyl record manufacturing system **20** will now be explained.

(69) The heating system **66** is provided to heat the die assemblies **51**, **52** such that polymeric puck **24** (shown in FIG. **2**) is compressible and in printable between the die assemblies **51**, **52**. The heating system **66** is fluidly coupled to the inlet aperture **168** and the outlet aperture **176** of the first die assembly **51**. Further, the heating system **66** is fluidly coupled to the inlet aperture **652** and the outlet aperture **654** of the second die assembly **52**. During operation, the heating system **66** pumps steam through the flow path of the first die assembly **51** including the inlet aperture **168**, the first vertical aperture **172**, the plurality of concentric channels **416**, the plurality of radial passages **424**,

the second vertical aperture **180**, and the outlet aperture **176**. The heating system **66** pumps the steam through the flow path of the first die assembly **51** in response to a control signal received from the machine controller **86**. Further, the heating system **66** pumps steam through the flow path of the second die assembly **52** in response to the control signal received from the machine controller **86**.

(70) The cooling system **70** is provided to cool the die assemblies **51**, **52** such that vinyl record **28** can be removed from the die assemblies **51**, **52**. The cooling system **70** is fluidly coupled to the inlet aperture **168** and the outlet aperture **176** of the first die assembly **51**. Further, the cooling system **70** is fluidly coupled to the inlet aperture **652** and the outlet aperture **654** of the second die assembly **52**. During operation, the cooling system **70** pumps a cooled fluid through the flow path of the first die assembly **51** including the inlet aperture **168**, the first vertical aperture **172**, the plurality of concentric channels **416**, the plurality of radial passages **424**, the second vertical aperture **180**, and the outlet aperture **176**. The heating system **66** pumps the steam through the flow path of the first die assembly **51** in response to a control signal received from the machine controller **86**. Further, the cooling system **70** pumps a cooled fluid through the flow path of the second die assembly **52** in response to the control signal received from the machine controller **86**.

(71) The actuator **81** is operably coupled to the first die assembly **51**. The actuator **81** moves the first die assembly **51** in a first direction (upwardly in FIG. **1**) in response to a control signal from the machine controller **86**. Further, the actuator **81** moves the first die assembly **51** in a second direction (downwardly in FIG. **1**) in response to another control signal from the machine controller **86**. The second die assembly **52** is held at a fixed position.

(72) The machine controller **86** is provided to control operation of the vinyl record manufacturing system **20**. The machine controller **86** is operably coupled to the heating system **66**, the cooling system **70**, and the actuator **81**. The machine controller **86** executes software instructions to control operation of the heating system **66**, the cooling system **70**, and the actuator **81** to transform the polymeric puck **24** into the vinyl record **28**.

(73) Referring to FIGS. **1** and **35-36**, a flowchart of a method for forming the vinyl record **28** utilizing the vinyl record manufacturing system **20** in accordance with another exemplary embodiment will now be explained.

(74) At step **700**, a user provides a vinyl record manufacturing system **20** (shown in FIG. **1**) having a first die assembly **51**, a second die assembly **52**, a first stamper plate **61** coupled to the first die assembly **51**, and a second stamper plate **62** coupled to the second die assembly **52**, a heating system **66**, a cooling system **70**, and at least one actuator **81**. The first die assembly **51** includes a base circular-shaped plate **100** (shown in FIGS. **14-13**) having a primary surface **140** and an outer ring-shaped gasket groove **152**, an inner O-ring groove **156**, and a plurality of concentric grooves **160** extending from the primary surface **140** into the base circular-shaped plate **100**. The plurality of concentric grooves **160** are disposed between the outer ring-shaped gasket groove **152** and the inner O-ring groove **156**. The base circular-shaped plate **100** having an inlet aperture **168** extending from an outer peripheral surface **144** thereof into the base circular-shaped plate **100** to a central region **184** of the base circular-shaped plate **100**. The inlet aperture **168** communicates with a first vertical aperture **172** that extends from an end of the inlet aperture **168** and through the primary surface **140**. The base circular-shaped plate **100** has an outlet aperture **176** extending from the outer peripheral surface **144** into the base circular-shaped plate **100**. The outlet aperture **176** communicates with a second vertical aperture **180** that extends from an end of the outlet aperture **176** and through the primary surface **140**. The base circular-shaped plate **100** having a plurality of peripheral bolt apertures **188** extending therethrough that are disposed radially outwardly from the outer ring-shaped gasket groove **152**. The first die assembly **51** further includes an outer ring-shaped gasket **105** (shown in FIG. **14**) disposed in the outer ring-shaped gasket groove **152**. The first die assembly **51** further includes an O-ring **108** (shown in FIGS. **5** and **14**) that is disposed in the inner O-ring groove **156**. The first die assembly **51** further includes a contacting circular-shaped

plate **112** (shown in FIGS. **24-31**) having an engagement surface **400** and a plurality of concentric channels **416** extending from the engagement surface **400** into the contacting circular-shaped plate **112** such that a plurality of concentric finger portions **420** are formed on the contacting circular-shaped plate **112**. Each concentric finger portion is disposed between two concentric channels of the plurality of concentric channels **416**. The contacting circular-shaped plate **112** has a plurality of radial passages **424** fluidly interconnecting the plurality of concentric channels **416**. The contacting circular-shaped plate **112** is disposed on the base circular-shaped plate **100** such that the plurality of concentric finger portions **420** are partially disposed in the plurality of concentric grooves **160** and contact the base circular-shaped plate **100**. The inlet aperture **168** fluidly communicates with the plurality of concentric channels **416**. The outlet aperture **176** fluidly communicates with the plurality of concentric channels **416**. A flow path is formed including the inlet aperture **168**, the first vertical aperture **172**, the plurality of concentric channels **416**, the plurality of radial passages **424**, the second vertical aperture **180**, and the outlet aperture **176**. The contacting circular-shaped plate **112** has a plurality of peripheral bolt holes **188** extending therein that are disposed radially outwardly from the plurality of concentric channels **416**. The first die assembly **51** further includes a plurality of bolts **122** extending through the plurality of peripheral bolt apertures **188** of the base circular-shaped plate **100** and further into the plurality of peripheral bolt holes **428** of the contacting circular-shaped plate **112** to couple the base circular-shaped plate **100** to the contacting circular-shaped plate **112**.
(75) At step **702**, the user disposes a polymeric puck **24** (shown in FIG. **2**) between the first and second stamper plates **61**, **62**.
(76) At step **704**, the heating system **66** (shown in FIG. **1**) heats the first and second die assemblies **51**, **52** by pumping steam through the flow path of the first die assembly **51** and through a flow path in the second die assembly **52**.
(77) At step **706**, the actuator **81** (shown in FIG. **1**) moves the first die assembly **51** towards the second die assembly **52** such that the first and second stamper plates **61**, **62** compress the polymeric puck **24** to form the vinyl record **28** (shown in FIG. **3**).
(78) At step **708**, the cooling system **70** (shown in FIG. **1**) cools the first and second die assemblies **51**, **52** by pumping a cooled fluid through the flow path of the first die assembly **51** and through a flow path in the second die assembly **52**.
(79) At step **710**, the actuator **81** (shown in FIG. **1**) moves the first die assembly **51** away from the second die assembly.
(80) At step **712**, the user removes the vinyl record **28** from at least one of the first and second stamper plates **61**, **62**.
(81) While the claimed invention has been described in detail in connection with only a limited number of embodiments, it should be readily understood that the invention is not limited to such disclosed embodiments. Rather, the claimed invention can be modified to incorporate any number of variations, alterations, substitutions or equivalent arrangements not heretofore described, but which are commensurate with the spirit and scope of the invention. Additionally, while various embodiments of the claimed invention have been described, it is to be understood that aspects of the invention may include only some of the described embodiments. Accordingly, the claimed invention is not to be seen as limited by the foregoing description.

Claims

1. A die assembly, comprising: a base circular-shaped plate having a primary surface and an outer ring-shaped gasket groove extending from the primary surface into the base circular-shaped plate, an inner O-ring groove extending from the primary surface into the base circular-shaped plate, and a plurality of concentric grooves extending from the primary surface into the base circular-shaped plate and being disposed between the outer ring-shaped gasket groove and the inner O-ring groove,

the base circular-shaped plate having an inlet aperture extending from an outer peripheral surface thereof into the base circular-shaped plate to a central region of the base circular-shaped plate, the inlet aperture communicating with a first vertical aperture that extends from an end of the inlet aperture and through the primary surface, the base circular-shaped plate having an outlet aperture extending from the outer peripheral surface into the base circular-shaped plate, the outlet aperture communicating with a second vertical aperture that extends from an end of the outlet aperture and through the primary surface; a contacting circular-shaped plate configured to be coupled to the base circular-shaped plate, the contacting circular-shaped plate having an engagement surface and a plurality of concentric channels extending from the engagement surface into the contacting circular-shaped plate such that a plurality of concentric finger portions is formed on the contacting circular-shaped plate, each concentric finger portion being disposed between two concentric channels of the plurality of concentric channels, the contacting circular-shaped plate having a plurality of radial passages fluidly interconnecting the plurality of concentric channels, the contacting circular-shaped plate, when coupled with the base circular-shaped plate, is disposed on the base circular-shaped plate such that each of the plurality of concentric finger portions contact the base circular-shaped plate and each is partially received by and disposed within respective one of the plurality of concentric grooves of the base circular-shaped plate, the inlet aperture fluidly communicating with the plurality of concentric channels, the outlet aperture fluidly communicating with the plurality of concentric channels, and a flow path being formed by and through the inlet aperture, the first vertical aperture, the plurality of concentric channels, the plurality of radial passages, the second vertical aperture, and the outlet aperture; an outer ring-shaped gasket that, when the contacting circular-shaped plate is coupled with the base circular-shaped plate, is disposed in the outer ring-shaped gasket groove of the base circular-shaped plate and contact the contacting circular-shaped plate, wherein the outer ring-shaped gasket groove is configured to have a depth greater than a depth of each of the plurality of concentric grooves of the base circular-shaped plate; and an O-ring that, when the contacting circular-shaped plate is coupled with the base circular-shaped plate, is disposed in the inner O-ring groove of the base circular-shaped plate and contact the contacting circular-shaped plate.

2. The die assembly of claim 1, wherein; a radial width of the outer ring-shaped gasket groove being greater than a radial width of the inner O-ring groove.

3. The die assembly of claim 1, wherein: the base circular-shaped plate having a center aperture extending therethrough; the contacting circular-shaped plate having a center aperture extending therethrough; and a threaded bolt being threadably disposed in the center aperture of the base circular-shaped plate and the center aperture of the contacting circular-shaped plate.

4. The die assembly of claim 1, wherein: the base circular-shaped plate having a ring-shaped placement groove extending from the primary surface into the base circular-shaped plate, the ring-shaped placement groove extending concentrically around the outer ring-shaped gasket groove, a depth of the ring-shaped placement groove being greater than the depth of the outer ring-shaped gasket groove, the depth of the ring-shaped placement groove being greater than the of the plurality of concentric grooves, a radial width of the ring-shaped placement groove being greater than a radial width of the outer ring-shaped gasket groove; and the contacting circular-shaped plate having a ring-shaped placement member extending outwardly from the engagement surface and around an outer periphery of the engagement surface, the ring-shaped placement member being received in the ring-shaped placement groove of the base circular-shaped plate.

5. The die assembly of claim 4, wherein: the ring-shaped placement member of the contacting circular-shaped plate having first and second cut-out regions extending therethrough.

6. The die assembly of claim 1, wherein: the contacting circular-shaped plate having an outer peripheral surface with first, second, third, and fourth clamping grooves disposed therein; and the outer peripheral surface of the base circular-shaped plate having first, second, third, and fourth clamping grooves disposed therein, that are aligned with the first, second, third, and fourth

clamping grooves, respectively, of the contacting circular-shaped plate.
7. The die assembly of claim 1, wherein: the central region of the base circular-shaped plate being disposed between the inner O-ring groove and a first concentric groove of the plurality of concentric grooves disposed radially outwardly from the inner O-ring groove.
8. The die assembly of claim 7, wherein: the first vertical aperture extending through the central region of the base circular-shaped plate and the primary surface.
9. The die assembly of claim 1, wherein: the second vertical aperture of the base circular-shaped plate communicates with the radially outermost concentric channel of the plurality of concentric channels.
10. The die assembly of claim 1, further comprising: a stamper plate being disposed on a stamper plate holding surface of the contacting circular-shaped plate, the stamper plate holding surface being disposed opposite to the engagement surface.
11. A method of forming a vinyl record, comprising: providing a vinyl record manufacturing system having a first die assembly, a second die assembly, a first stamper plate coupled to the first die assembly, and a second stamper plate coupled to the second die assembly, the first die assembly including a base circular-shaped plate having a primary surface and an outer ring-shaped gasket groove extending from the primary surface into the base circular-shaped plate, an inner O-ring groove extending from the primary surface into the base circular-shaped plate, and a plurality of concentric grooves extending from the primary surface into the base circular-shaped plate and being disposed between the outer ring-shaped gasket groove and the inner O-ring groove, the base circular-shaped plate having an inlet aperture extending from an outer peripheral surface thereof into the base circular-shaped plate to a central region of the base circular-shaped plate, the inlet aperture communicating with a first vertical aperture that extends from an end of the inlet aperture and through the primary surface, the base circular-shaped plate having an outlet aperture extending from the outer peripheral surface into the base circular-shaped plate, the outlet aperture communicating with a second vertical aperture that extends from an end of the outlet aperture and through the primary surface, the base circular-shaped plate having a plurality of peripheral bolt apertures extending therethrough that are disposed radially outwardly from the outer ring-shaped gasket groove, the first die assembly further including an outer ring-shaped gasket being disposed in the outer ring-shaped gasket groove, wherein the outer ring-shaped gasket groove is configured to have a depth greater than a depth of each of the plurality of concentric grooves of the base circular-shaped plate, the first die assembly further including an O-ring being disposed in the inner O-ring groove, the first die assembly further including a contacting circular-shaped plate having an engagement surface and a plurality of concentric channels extending from the engagement surface into the contacting circular-shaped plate such that a plurality of concentric finger portions is formed on the contacting circular-shaped plate, each concentric finger portion being disposed between two concentric channels of the plurality of concentric channels, the contacting circular-shaped plate having a plurality of radial passages fluidly interconnecting the plurality of concentric channels, the contacting circular-shaped plate being disposed on the base circular-shaped plate such that each of the plurality of concentric finger portions contact the base circular-shaped plate and each is partially received by and disposed within respective one of the plurality of concentric grooves of the base circular-shaped plate, the inlet aperture fluidly communicating with the plurality of concentric channels, the outlet aperture fluidly communicating with the plurality of concentric channels, and a flow path being formed by and through the inlet aperture, the first vertical aperture, the plurality of concentric channels, the plurality of radial passages, the second vertical aperture, and the outlet aperture, the contacting circular-shaped plate having a plurality of peripheral bolt holes extending therein that are disposed radially outwardly from the plurality of concentric channels, the first die assembly further including a plurality of bolts extending through the plurality of peripheral bolt apertures of the base circular-shaped plate and further into the plurality of peripheral bolt holes of the contacting circular-shaped plate to couple the base circular-shaped plate

to the contacting circular-shaped plate; disposing a polymeric puck between the first and second stamper plates; heating the first and second die assemblies by pumping steam through the flow path of the first die assembly and through a flow path in the second die assembly; and moving the first and second die assemblies toward one another such that the first and second stamper plates compress the polymeric puck to form the vinyl record.

12. The method of claim 11, further comprising: cooling the first and second die assemblies by pumping a cooled fluid through the flow path of the first die assembly and through the flow path in the second die assembly; moving the first and second die assemblies away from one another; and removing the vinyl record from at least one of the first and second stamper plates.